- When a model you rely on dictates that only a few specific constraints (like mean and variance) need to hold true, how do you prevent yourself from falsely assuming that all other hidden variables within that system are identical?

User prompt: What should I probably include on my A4 paper of notes? Highlight top 5 cookie cutter methods that'll probably be on the exam in 2026

Response: Because the course provides the distribution tables and characteristic function formulary directly on the exam, reserve your single handwritten A4 sheet exclusively for multivariate formulas, limit theorems, conditioning tricks, and transformation templates.

Top 5 Cookie-Cutter Methods for the Exam

1. Bivariate / Multivariate Transformation via Jacobian

- **When to use:** Given joint density $f_{X,Y}(x,y)$, find $f_{U,V}(u,v)$ where $U = g_1(X,Y)$ and $V = g_2(X,Y)$.
- **The Blueprint:**
  1. Invert the equations: express $x = h_1(u,v)$ and $y = h_2(u,v)$.
  2. Compute the Jacobian matrix determinant:

$$J = \det \begin{pmatrix} \frac{\partial h_1}{\partial u} & \frac{\partial h_1}{\partial v} \\ \frac{\partial h_2}{\partial u} & \frac{\partial h_2}{\partial v} \end{pmatrix}, \quad |J| = |\det(J)| [\text{cite: 4, 17}]$$

  3. Plug into density formula: $f_{U,V}(u,v) = f_{X,Y}(h_1(u,v), h_2(u,v)) \cdot |J|$.
  4. Carefully map the support inequalities into (u,v)-space.
  5. Test independence: check if $f_{U,V}(u,v) = f_U(u) f_V(v)$ on a product domain $I_u \times I_v$.

2. Multivariate Normal: Conditioning & Linear Transformations

- **When to use:** Linear combinations $Z = BX$ of a normal vector, or computing $\mathbb{E}[Z_1 \mid Z_2]$ and $Var(Z_1 \mid Z_2)$.
- **The Blueprint:**
  - **Linear transformation:** If $X \sim \mathcal{N}(\mu, \Sigma)$, then $Z = BX \sim \mathcal{N}(B\mu, B\Sigma B')$.
  - **Bivariate conditional normal formula:**

$$\mathbb{E}[Z_1 \mid Z_2 = z_2] = \mathbb{E}[Z_1] + \frac{\text{Cov}(Z_1, Z_2)}{\text{Var}(Z_2)}(z_2 - \mathbb{E}[Z_2]) [\text{cite: 18}]$$

$$\mathrm{Var}(Z_1 \mid Z_2 = z_2) = \mathrm{Var}(Z_1) - \frac{\mathrm{Cov}(Z_1, Z_2)^2}{\mathrm{Var}(Z_2)} = \mathrm{Var}(Z_1)(1-\rho^2)[\text{cite: 18}]$$

- **Quadratic form matching:** If $f_X(x) \propto \exp(-\frac{1}{2}x'Ax)$, then $\Sigma = A^{-1}$ and μ is derived from linear terms.
- **Independence:** For jointly normal variables, $\mathrm{Cov}(X_i, X_j) = 0 \iff$ independence.

3. Conditioning via Law of Total Expectation / Probability

- **When to use:** Compound sums ($S_N = \sum_{j=1}^{N} X_j$), random stops, or finding characteristic functions of mixed models.
- **The Blueprint:**
  - $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \sum_k \mathbb{E}[Y \mid X = k]\mathbb{P}(X = k)$ (or $\int \mathbb{E}[Y \mid X = x] f_X(x) dx$).
  - **Conditional density:** $f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)}$ where $f_X(x) = \int f_{X,Y}(x,y)dy$.
  - **Truncated expectations:** $\mathbb{E}[Y \mid A] = \frac{\mathbb{E}[Y\mathbf{1}_A]}{\mathbb{P}(A)}$.
  - **Random sum identities:**

$$F_{Z_N}(z) = g_N(F_{X_1}(z)) \quad \text{where } g_N(t) = \mathbb{E}[t^N][\text{cite: 11}]$$

$$\mathbb{E}[S_N] = \mathbb{E}[N]\mathbb{E}[X_1] \quad (\text{Wald's identity, assuming independence})$$

4. Characteristic Function Properties & Limits (Continuity Theorem)

- **When to use:** Establishing convergence in distribution, symmetry, or distributions of independent sums/differences.
- **The Blueprint:**
  - $\varphi_{aX+b}(t) = e^{itb}\varphi_X(at)$.
  - $\varphi_{-X}(t) = \varphi_X(-t) = \overline{\varphi_X(t)}$.
  - X is symmetric ($X \overset{d}{=} -X$) $\iff$ $\varphi_X(t) \in \mathbb{R}$ for all t.
  - If X, Y are i.i.d., then $\varphi_{X-Y}(t) = |\varphi_X(t)|^2 \geq 0$.
  - **Continuity Theorem:** If $\varphi_{X_n}(t) \to \varphi_X(t)$ pointwise and the limit is continuous at $t = 0$, then $X_n \xrightarrow{d} X$.

- **Taylor expansion around** 0: $\varphi_X(t) = 1 + it\mathbb{E}[X] - \frac{t^2}{2}\mathbb{E}[X^2] + o(t^2)$.

5. Order Statistics and Modes of Convergence

- **When to use:** Maximums/minimums ($Z_n = \max_{1\le k\le n} X_k$), showing convergence in probability or distribution.
- **The Blueprint:**
  - **Max / Min distribution functions:**

$$F_{\max}(z) = \prod_{k=1}^{n} F_{X_k}(z) \overset{\text{i.i.d.}}{=} (F_{X_1}(z))^n [\text{cite: 10}]$$

$$F_{\min}(z) = 1 - \prod_{k=1}^{n} (1 - F_{X_k}(z)) \overset{\text{i.i.d.}}{=} 1 - (1 - F_{X_1}(z))^n$$

  - **Convergence in distribution to constant:** If $F_{X_n}(x) \to F_c(x)$ for all continuity points of F_c, then $X_n \xrightarrow{d} c \iff X_n \xrightarrow{p} c$.
  - **Inequalities for bounding:**
    - Markov: $\mathbb{P}(|X| \ge a) \le \frac{\mathbb{E}[|X|]}{a}$
    - Chebyshev: $\mathbb{P}(|X - \mu| \ge a) \le \frac{\text{Var}(X)}{a^2}$

What Else to Squeeze Onto the Sheet

- **Standard geometric series derivatives:** $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$, $\sum_{n=1}^{\infty} nr^{n-1} = \frac{1}{(1-r)^2}$, $\sum_{n=2}^{\infty} n(n-1)r^{n-2} = \frac{2}{(1-r)^3}$.
- **Standard Taylor expansions:** $e^u = 1 + u + \frac{u^2}{2} + O(u^3)$, $\log(1+u) = u - \frac{u^2}{2} + O(u^3)$.
- **Delta Method statement:** If $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$, then $\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, \sigma^2[g'(\theta)]^2)$.
- **Slutsky's Theorem:** If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$, $X_n Y_n \xrightarrow{d} cX$, and $X_n / Y_n \xrightarrow{d} X/c$ (if $c \neq 0$).
- In your preparation timeline, are you dedicating time to practice reconstructing joint integration bounds from inequalities under strict exam-style time limits, or are you primarily focusing on memorizing formulas?